

11th and 12th Week Summary (11/19/26)

- **ANOVA:** If the assumption of the equality of variances fails
- We “transform” the data. The common transformations are
 - If $\sigma^2 \propto \mu$, the use $Y_T = \sqrt{Y}$ or $\sqrt{Y + 0.375}$
 - If $\sigma^2 \propto \mu^2$, the use $Y_T = \ln(Y)$ or $\ln(Y + 1)$
 - If $\sigma^2 \propto \mu(1 - \mu)$, the use $Y_T = \sin^{-1} \sqrt{Y}$
- Assumptions for Oneway ANOVA model ($y_{ij} = \mu + \tau_i + \epsilon_{ij}$, where $i = 1, \dots, t$ and $j = 1, \dots, n_i$):
 - To test $H_0 : \tau_i = 0$ vs $H_a : \tau_i \neq 0$ for some i :
 - (1) The ϵ_{ij} 's are independent and normally distributed
 - (2) $\text{Var}(\epsilon_{ij}) = \sigma^2$ (a constant value)
- Checking the Assumptions:
 - Obtain the residuals ($r_{ij} = y_{ij} - \hat{\mu} - \hat{\tau}_i$) and fitted values ($\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i$), then
 - (1) The QQ-plot of r_{ij} 's should be linear
 - (2) The scatterplot of r_{ij} 's versus $\hat{y}_{ij}$'s should follow a random pattern.
- **Multiple Comparisons**
- The problem of **multiplicity** is serious when we are testing many hypotheses
- Instead of using Type-I error rate, we use familywise error rate (FEW):
 - $\alpha_F = P(\text{Falsely reject at least one hypotheses})$
- Bonferroni Method:
 - If there are m hypotheses, then Reject $H_0^{ij} : \mu_i = \mu_j$ if $|\bar{y}_i - \bar{y}_j| > t_{\alpha/2m} \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$
 - Pros:** This will guarantee that $\alpha_F \leq 0.05$
 - Cons:** The chance of rejecting H_0 is small. In other word, the power of Bonferroni is very poor
 - R: with(exmp9.3, pairwise.t.test(yield, agent, p.adj = "bonf"))
- Fisher's Least Significant Difference (LSD):
 - Reject $H_0^{ij} : \mu_i = \mu_j$ if $|\bar{y}_i - \bar{y}_j| > t_{\alpha/2} \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$
 - Pros:** Easy to discover differences (reject H_0^{ij}). In other word, high power
 - Cons:** This DOES NOT control the familywise error
 - R: with(exmp9.3, pairwise.t.test(yield, agent, p.adj = "none"))
- Tukey's Method:
 - Reject $H_0^{ij} : \mu_i = \mu_j$ if $|\bar{y}_i - \bar{y}_j| > \frac{q_\alpha(t, df_E)}{\sqrt{2}} \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$
 - Pros:** Control the familywise error rate
 - Cons:** Lower power comparing to Fisher LSD and Dunnett methods
 - R: model = aov(yield agent, data= exmp9.3); TukeyHSD(model)
- Dunnett's Method:
 - Reject $H_0^{ij} : \mu_i = \mu_j$ if $|\bar{y}_i - \bar{y}_j| > d_\alpha(t - 1, df_E) \sqrt{\frac{2MSE}{n}}$
 - Pros:** Control the familywise error rate. Higher power comparing to Bonferroni and Tukey's
 - Cons:** It's just for comparing with a control group.
 - R: library("DescTools"); with(exmp9.3, DunnettTest(yield, agent, control = "None"))